Chapter 13: The New Keynesian Framework

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1 The New Keynesian Model

1.1 Overview: Effect of an anticipated monetary expansion

While the New Classical Phillips curve helps explaining the breakdown of the traditional Phillips Curve in the 1970's, it has predictions that is difficult to square with evidence. A key predictions is that a monetary expansion only has an effect in the period where it is not anticipated. The evidence, however, suggest that in response to a surprise cut in the Federal Funds Rate, output responds with a considerable lag and peaks in 6 quarters. This means that as a result of a policy shock there is a forecastable increase in output over a considerable period.

Let us first summarize the evidence on the effect of monetary policy through its effect on nominal and real GDP response to monetary policy shock. The monetary shock is defined as unanticipated cut in the Federal Funds rate. Figure 1 shows that in response to a monetary policy shock nominal GDP increases permanently. Figure 2 shows the effect on real GDP. It increases temporarily with a peak in 6 quarters. Both figures are from Woodford (2003)

The nice thing about summarizing this evidence in terms of nominal and real GDP is that if we summarize aggregate demand as an exogenous path for nominal GDP, then we can "close" a model by only specifying a Phillips curve. This allows us to isolate the implications of the Phillips curve, without needing to be explicit about the demand side of the model (i.e. the IS curve and the assumed policy rule) as I show below.

Let us define nominal demand as $D_t \equiv P_t Y_t$ Suppose that nominal demand follows the exogenous process

$$\Delta \hat{D}_t = \rho \Delta \hat{D}_{t-1} + \epsilon_t + q_t, \tag{1}$$

where ϵ_t is a monetary policy shock and q_t is any other shock perturbing the economy, which we assume is not correlated to monetary policy shocks. You

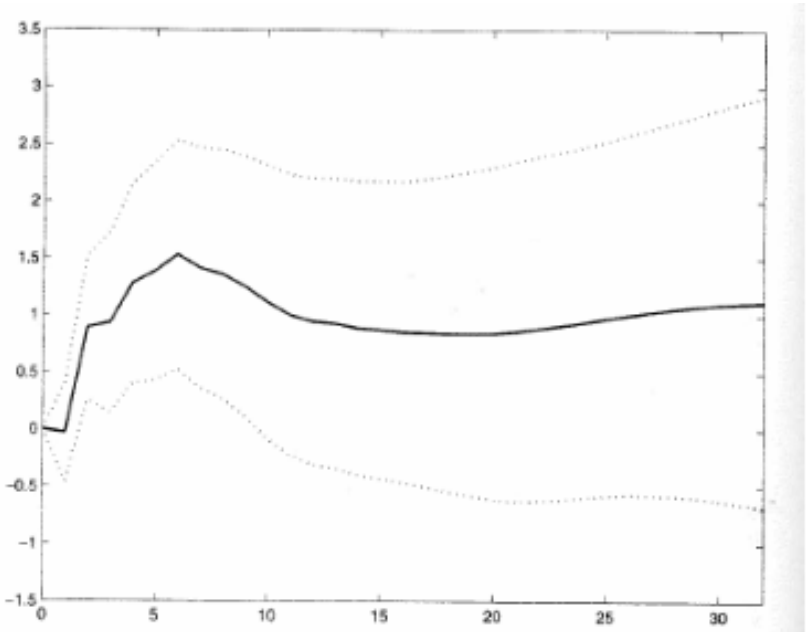


Figure 3.2 *Impulse response of nominal GDP to an unanticipated interest-rate reduction in quarter zero. Periods on horizontal axis represent quarters and the vertical axis measures the effect on log nominal GDP in percentage points. Source: Christiano et al. (2001).*

Figure 1: Effect of Monetary Shock on Nominal GDP

should think of the monetary policy and fiscal policy as having been implicitly chosen so as to generate this process. This process captures the qualitative response of nominal GDP in response to a monetary policy shock. The impulse response of nominal GDP in response to an innovation ϵ_t is shown by a dashed line in figure 3. A Phillips curve, then, should be able to capture a humps shaped response for real GDP in response to ϵ_t, and let me make it more explicit why the New Classical Phillips curve is bound to fail.

Consider the linearized version of the New Classical Phillips curve derived in last chapter:

$$\pi_t = \kappa \hat{Y}_t + E_{t-1}\pi_t$$

and observe that output and inflation are related by:

$$\Delta \hat{D}_t = \hat{Y}_t - \hat{Y}_{t-1} + \pi_t \tag{2}$$

Consider a shock ϵ_t at time 0. What is the effect on real output? We see immediately from this Phillips curve that $E_0 \hat{Y}_{t+j} = 0$ for $j > 0$. Hence, the

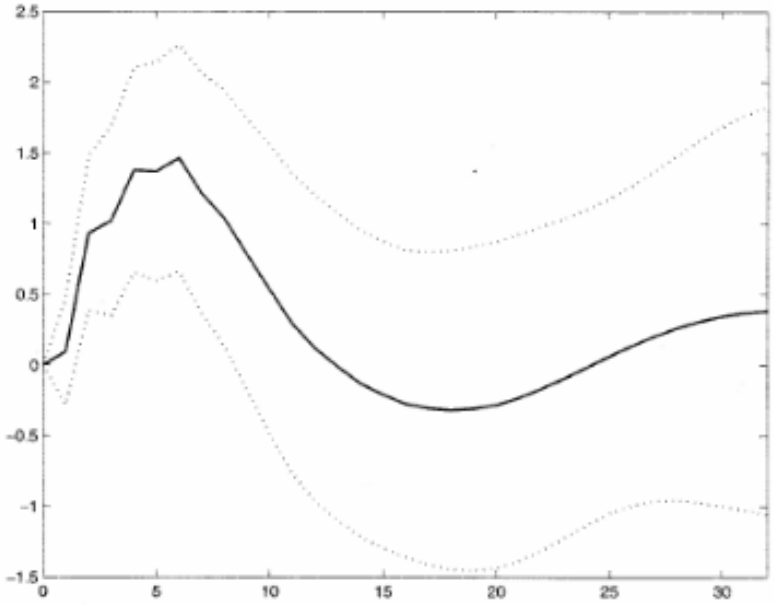


Figure 3.3 *Impulse response of real GDP to the same monetary policy shock as in Figure 3.2. The vertical axis measures the effect on log real GDP in percentage points. Source: Christiano et al. (2001).*

Figure 2: Effect of Monetary Shock on Real GDP

impulse response for output at time $t > 0$ is flat at zero and the model cannot replicate the empirical impulse response which implies a forecastable increase in real output as a result of the monetary expansion.

Consider instead the New Keynesian Phillips curve, which we will derive in this chapter:

$$\pi_t = \kappa \hat{Y}_t + \beta E_t \pi_{t+1} \tag{3}$$

Solving (1)-(3) together, we can compute the implied impulse responses of inflation and real output. As shown by the black solid line in figure 3, taken from Woodford (2003), the New Keynesian Phillips curve capture qualitatively the hump-shaped behavior of real output in response to innovation in nominal demand as seen in the empirical impulse response in figure 2.

A central feature of this model is that even if all firms set their price optimally, they do so in a *staggered fashion*, that is, at any given time, only a few firms set their price. Why is this important? Because in this case, since only a few firms set their price, then they may not have incentive to set their prices too differently from other firms, for the same reason we analyzed in the context

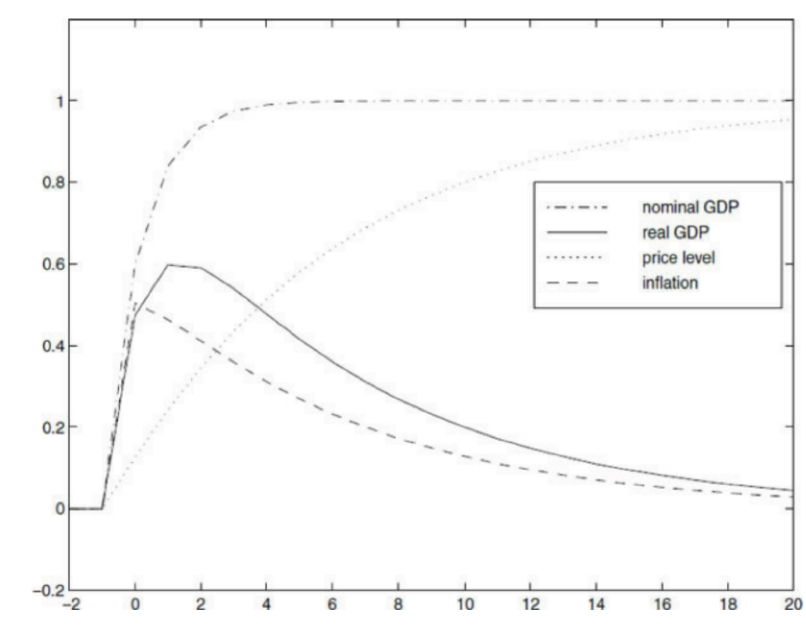


Figure 3: Impulse Response of nominal and real GDP in response to a monetary policy shock according the the New Keynesian model

of the dynamic IS-LM model, i.e due to *real rigidities*.

Observe also that this Phillips curve also address the key problem with the standard Phillips curve and its breakdown discussed in last chapter. Here expectations about inflation also shift the trade-off between inflation and output, and thus the New Keynesian Phillips curve is consistent the type of evidence of "breakdown" that we showed.

We will now show how to derive equation (3) from microfoundations.

1.2 New Keynesian Phillips Curve

Each firms sets a new price for its product with probability $(1-\alpha)$, $0 < \alpha < 1$, in each period. This probability is independent from the time of last price change. α us tgerefire measures the degree of nominal rigidity. The expected time until the next price change is:

$$(1-\alpha)\cdot 0 + \alpha\cdot(1-\alpha)\cdot 1 + \alpha^2\cdot(1-\alpha)\cdot 2 + \cdots = \frac{\alpha}{1-\alpha}$$

There are several nice properties of this assumption:

i. We have a one parameter family of models that allows for arbitrary variation in average time until price changes between zero and infinity. Meanwhile the form of the resulting difference equations (3) is the same, as we will see, regardless of the assumed degree of price rigidity, with only κ being affected;

ii. Assuming that the probability of price change is independent from the time since last price change reduces the relevant number of state variables: we don't need to keep track of how existing prices vary with age of the price commitment (even if not realistic, it brings enormous gains in tractability). Moreover, it gives a very transparent way of modeling staggered prices.

Each firm hires labor from separate labor markets. Thus, we assume that the households are supplying several different labor "types"

$$\max_{C_t,M_t,B_t} E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}} + \chi \frac{\left(\frac{M_t}{P_t}\right)^{1-\nu}}{1-\nu} - \int_0^1 \frac{N_t(i)^{1+\varphi}}{1+\varphi} di \right] \xi_t$$

s.t.

$$\int_0^1 p_t(i)c_t(i)di + M_t + B_t = (1+i_{t-1})B_{t-1} + M_{t-1} + \int_0^1 W_t(i)N_t(i)di - T_t + \int_0^1 Z_t(i)di$$

Relative to our earlier exposition, we now have a first order condition for every labor type:

$$\frac{W_t(i)}{P_t} = C_t^{\sigma^{-1}} N_t(i)^{\varphi}$$

as here labor is being supplied to different labor markets.

Recall from our earlier derivation of Phillips curves, is that the derivation essentially boils down to two steps, first you find the price of all firms are setting at a given time, and then you substitute this into the price index.

We start by finding the optimal price for a firm that gets to reset its price at time t. The firms maximizes the net present value of profits:

$$E_t \sum_{T=t}^{\infty} Q_{t,T} Z_T(i)$$

In the Calvo model, as in the other models we have covered, it is assumed that the firm sets its price and then will hire labor to satisfy whatever is demanded at that price. Again, for simplicity, we consider a linear production function. We want to take the derivative of the profit function with respect to $p_t(i)$ at time to and we need to take into account the future elements of profits that

depend upon $p_t(i)$. For a firm that gets to set a price at time t, we can rewrite its profits to make clear which part of future profits depend on this price as:

$$\max_{p_t(i)} E_0 \left\{ \sum_{T=t}^{\infty} \alpha^{T-t} Q_{t,T} Z_T(p_t(i)) \right\} + \text{terms that do not depend upon } p_t(i)$$

where we are denoting profits at time T that depends on $p_t(i)$ by $Z_T(p_t(i))$ and observe that they are weighted by α^{T-t}.

To see that we can write the profit function in this way, consider profits at time $t+1$, conditional on prices having been changed at time t. At this time, there is a probability α of prices being unchanged and probability $1-\alpha$ that they are changed again. If prices are not changed, we evaluate the profit at time $t+1$ by probability α and take into account how these profits change with different value for $p_t(i)$. We do not have to take into account the terms conditional on the firms readjusting its price: the firm will be setting a new price and the old price has not effect on the profit at that time – this is the key benefit of the Calvo assumption. Hence, we can omit it. Hence, in order to find the optimal price chosen at time t, conditional on it being in effect with probability α in each period, the firm maximizes:

$$\max_{p_t(i)} E_0 \sum_{T=t}^{\infty} \alpha^{T-t} Q_{t,T} \left[p_t(i)^{1-\theta_t} P_T^{\theta_t} Y_T - W_T(i) p_T(i)^{-\theta_t} P_t^{\theta_t} \frac{Y_T}{A_T} \right]$$

resulting in the first order condition:

$$E_t \sum_{T=t}^{\infty} \alpha^{T-t} Q_{t,T} \left[(1-\theta_t) p_t(i)^{-\theta_t} P_T^{\theta_t} Y_T + \theta W_T(i) p_t(i)^{-\theta_t - 1} P_T^{\theta_t} \frac{Y_T}{A_T} \right] = 0$$

As this will apply to all firms setting their price at time t, their optimal price, call it p_t^*, satisfies (with a little bit of algebra):

$$E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma^{-1}} \left(\frac{p_t^*}{P_T} \right)^{-\theta_t} Y_T \xi_T \left(\frac{p_t^*}{P_T} - \frac{\theta_t}{\theta_t - 1} \frac{W_T^*}{P_T A_t} \right) = 0$$

where W_t^* is the wage rate prevailing in the market where the firm is hiring from and is given by:

$$\frac{W_t^*}{P_t} = C_t^{\sigma^{-1}} N_t^{*\varphi}$$

As production is linear in labor, then: $y_T^* = A_T N_T^*$, so that: $N_T^* = \frac{y_T^*}{A_T} = (\frac{p_t^*}{P_T})^{-\theta} \frac{Y_T}{A_T}$. Therefore:

$$\begin{aligned} \frac{W_T^*}{P_T} &= C_T^{\sigma^{-1}} N_T^{*\varphi} \\ &= C_T^{\sigma^{-1}} \left(\frac{p_t^*}{P_T} \right)^{-\theta\varphi} Y_T^{\varphi} A_T^{-\varphi} \end{aligned}$$

Substituting this into the FOC, a little bit of algebra (see Appendix) allows us to express this equation as:

$$\frac{p_t^*}{P_t} = \left(\frac{K_t}{F_t}\right)^{\frac{1}{1+\varphi\theta_t}} \tag{4}$$

where:

$$K_t \equiv \frac{\theta}{\theta_t - 1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \left(\frac{P_t}{P_T}\right)^{-\theta_t(1+\varphi)} \xi_T Y_T^{1+\varphi} A_T^{-\varphi-1} \tag{5}$$

$$F_t \equiv E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma^{-1}} \left(\frac{P_t}{P_T}\right)^{-\theta_t+1} Y_T \xi_T \tag{6}$$

The Calvo assumption implies that the price index can be expressed as:

$$P_t^{1-\theta_t} = (1-\alpha)(p_t^*)^{1-\theta_t} + \alpha P_{t-1}^{1-\theta_t} \tag{7}$$

Using (4) to substitute for $\frac{p_t^*}{P_t}$ in (7) we obtain:

$$\frac{K_t}{F_t} = \left(\frac{1-\alpha\Pi_t^{\theta_t-1}}{1-\alpha}\right)^{\frac{1+\theta_t\varphi}{1-\theta_t}} \tag{8}$$

It is convenient to express K_t and F_t in the following recursive form, which follow directly from the definition of these terms (with a little bit of algebra):

$$K_t = \frac{\theta_t}{\theta_t - 1} \xi_t Y_t^{1+\varphi} A_T^{-\varphi-1} + \alpha\beta E_t \Pi_{t+1}^{\theta(1+\varphi)} K_{t+1} \tag{9}$$

$$F_t = C_t^{-\sigma^{-1}} Y_t \xi_t + \alpha\beta E_t \Pi_{t+1}^{\theta-1} F_{t+1} \tag{10}$$

The aggregate supply is thus summarized by the three equations (8),(9) and (10) containing four endogenous variables K_t, F_t, Y_t, Π_t

The non-linear model can now be summarized as before with these three equations above replace the demand and supply for labor in our previous definitions. A linear approximation of these relationships give us the next proposition summarizing aggregate supply in the approximated equilibrium:

Proposition 1. *New Keynesian Phillips Curve. If a constant fraction α of firms reset their prices in each period, then the aggregate supply side of the approximated model summarized the New Keynesian curve*

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t \pi_{t+1} \tag{11}$$

where $\kappa = \frac{(1-\alpha)(1-\alpha\beta)}{\alpha} \frac{\varphi_\alpha + \sigma_c^{-1}}{1+\varphi_\alpha\theta}$.

We can write the model in terms of the efficient output gap, resulting in the IS and the AS equations:

$$x_t = E_t x_{t+1} - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{r}_t^e)$$

$$\pi_t = \kappa x_t + \beta E_t \pi_{t+1} + \omega\hat{\mu}_t$$

where $\omega \equiv \frac{\kappa}{\varphi_\alpha + \sigma_c^{-1}}$

The Calvo-Woodford model has a nice and well defined welfare function, derived by a second order approximation of the representative household, which takes the same form as the loss function with a standard Phillips curve. It is given by

$$\sum_{t=0}^{\infty} \beta^t \{\pi_t^2 + \lambda_x (x_t - x^*)^2\}$$

where $\lambda_x = \kappa/\theta$. This objective is derived under the assumption that real money balance contribute only trivially to utility (third order or lower). It now becomes interesting to think about optimal monetary policy, a topic we turn to next.

2 References

[1] Michael Woodford. *Interest and Prices. Foundations of a Theory of Monetary Policy*. Princeton University Press, 2003.

3 Appendix

3.1 How to get (4):

The first order condition is:

$$E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} \left(\frac{p_t^*}{P_T}\right)^{-\theta} Y_T \xi_T \left(\frac{p_t^*}{P_T} - \frac{\theta}{\theta - 1}\frac{W_T^*}{P_T A_t}\right) = 0$$

With:

$$\frac{W_T^*}{P_T} = C_T^{\sigma} \left(\frac{p_t^*}{P_T}\right)^{-\theta\varphi} Y_T^{\varphi} A_T^{-\varphi}$$

They can be combined to yield:

$$E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} \left(\frac{p_t^*}{P_T}\right)^{-\theta} Y_T \xi_T \left(\frac{p_t^*}{P_T} - \frac{\theta}{\theta - 1} C_T^{\sigma} \left(\frac{p_t^*}{P_T}\right)^{-\theta\varphi} Y_T^{\varphi} A_T^{-\varphi - 1}\right) = 0$$

Now take each piece of the summation to each side of the equation, simplify and collect terms, i.e.:

$$\begin{aligned} & E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} \left(\frac{p_t^*}{P_T}\right)^{-\theta+1} Y_T \xi_T \\ = \quad & \frac{\theta}{\theta-1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \left(\frac{p_t^*}{P_T}\right)^{-\theta(1+\varphi)} \xi_T Y_T^{1+\varphi} A_T^{-\varphi-1} \end{aligned}$$

We can take $(\frac{p_t^*}{P_t})^{-\theta+1}$ outside of the sum on the left hand side and $(\frac{p_t^*}{P_t})^{-\theta(1+\varphi)}$ on the right hand side:

$$\begin{aligned} & \left(\frac{p_t^*}{P_t}\right)^{-\theta+1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} \left(\frac{P_t}{P_T}\right)^{-\theta+1} Y_T \xi_T \\ = \quad & \left(\frac{p_t^*}{P_t}\right)^{-\theta(1+\varphi)} \frac{\theta}{\theta-1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \left(\frac{P_t}{P_T}\right)^{-\theta(1+\varphi)} \xi_T Y_T^{1+\varphi} A_T^{-\varphi-1} \end{aligned}$$

And now we get:

$$\left(\frac{p_t^*}{P_t}\right)^{1+\varphi\theta} = \frac{\frac{\theta}{\theta-1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} (\frac{P_t}{P_T})^{-\theta(1+\varphi)} \xi_T Y_T^{1+\varphi} A_T^{-\varphi-1}}{E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} (\frac{P_t}{P_T})^{-\theta+1} Y_T \xi_T}$$

or:

$$\frac{p_t^*}{P_t} = \left(\frac{K_t}{F_t}\right)^{\frac{1}{1+\varphi\theta}}$$

where:

$$\begin{aligned} K_t &\equiv \frac{\theta}{\theta-1} E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} \left(\frac{P_t}{P_T}\right)^{-\theta(1+\varphi)} \xi_T Y_T^{1+\varphi} A_T^{-\varphi-1} \\ F_t &\equiv E_t \sum_{T=t}^{\infty} (\alpha\beta)^{T-t} C_T^{-\sigma} \left(\frac{P_t}{P_T}\right)^{-\theta+1} Y_T \xi_T \end{aligned}$$

3.2 How to get (11):

We start by linearizing (9). Note that, since $\bar{\Pi} = 1$, (9) in steady state is:

$$\begin{aligned} \bar{K} &= \frac{\theta}{\theta-1} \bar{\xi} \bar{Y}^{1+\phi} \bar{A}^{-\phi-1} + \alpha\beta\bar{K} \\ \bar{K} &= \frac{\theta}{\theta-1} \frac{\bar{\xi} \bar{Y}^{1+\phi} \bar{A}^{-\phi-1}}{1-\alpha\beta} \end{aligned}$$

Now, we can start the linearization:

$$\begin{aligned} dK_t &= \frac{\theta}{\theta-1}\bar{Y}^{1+\phi}\bar{A}^{-\phi-1}d\xi_t + (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{\phi}\bar{A}^{-\phi-1}dY_t - (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{1+\phi} \\ &\quad \bar{A}^{-\phi-2}dA_t + \theta(1+\phi)\alpha\beta\bar{\Pi}^{\theta(1+\phi)-1}\bar{K}E_t d\Pi_{t+1} + \alpha\beta\bar{\Pi}^{\theta(1+\phi)}E_t dK_{t+1} \\ \bar{K}\hat{K}_t &= \frac{\theta}{\theta-1}\bar{Y}^{1+\phi}\bar{A}^{-\phi-1}\bar{\xi}\hat{\xi}_t + (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{\phi+1}\bar{A}^{-\phi-1}\hat{Y}_t - (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{1+\phi} \\ &\quad \bar{A}^{-\phi-1}\hat{A}_t + \theta(1+\phi)\alpha\beta\bar{\Pi}^{\theta(1+\phi)}\bar{K}E_t\pi_{t+1} + \alpha\beta\bar{\Pi}^{\theta(1+\phi)}\bar{K}E_t\hat{K}_{t+1} \end{aligned}$$

Since $\bar{\Pi} = 1$:

$$\begin{aligned} \bar{K}\hat{K}_t &= \frac{\theta}{\theta-1}\bar{Y}^{1+\phi}\bar{A}^{-\phi-1}\bar{\xi}\hat{\xi}_t + (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{\phi+1}\bar{A}^{-\phi-1}\hat{Y}_t - (1+\phi)\frac{\theta}{\theta-1}\bar{\xi}\bar{Y}^{1+\phi} \\ &\quad \bar{A}^{-\phi-1}\hat{A}_t + \theta(1+\phi)\alpha\beta\bar{K}E_t\pi_{t+1} + \alpha\beta\bar{K}E_t\hat{K}_{t+1} \end{aligned}$$

Divide both sides by $\bar{K}$ and obtain:

$$\hat{K}_t = (1-\alpha\beta)[\hat{\xi}_t + (1+\phi)\hat{Y}_t - (1+\phi)\hat{A}_t] + \alpha\beta[\theta(1+\phi)E_t\pi_{t+1} + E_t K_{t+1}] \quad (12)$$

Let's turn to equation (10). As before, in steady state we have:

$$\bar{F} = \bar{C}^{-\sigma}\bar{Y}\bar{\xi} + \alpha\beta\bar{F}$$
$$\bar{F} = \frac{\bar{C}^{-\sigma}\bar{Y}\bar{\xi}}{1-\alpha\beta}$$

Continue with the linearization:

$$\begin{aligned} dF_t &= -\sigma\bar{C}^{-\sigma-1}\bar{Y}\bar{\xi}dC_t + \bar{C}^{-\sigma}\bar{\xi}dY_t + \bar{C}^{-\sigma}\bar{Y}d\xi_t + (\theta-1)\alpha\beta\bar{\Pi}^{\theta-2}\bar{F}E_t d\Pi_{t+1} \\ &\quad +\alpha\beta\bar{\Pi}^{\theta-1}E_t dF_{t+1} \\ \bar{F}\hat{F}_t &= -\sigma\bar{C}^{-\sigma}\bar{Y}\bar{\xi}\hat{C}_t + \bar{C}^{-\sigma}\bar{\xi}\bar{Y}\hat{Y}_t + \bar{C}^{-\sigma}\bar{Y}\bar{\xi}\hat{\xi}_t + (\theta-1)\alpha\beta\bar{\Pi}^{\theta-1}\bar{F}E_t\pi_{t+1} \\ &\quad +\alpha\beta\bar{\Pi}^{\theta-1}\bar{F}E_t\hat{F}_{t+1} \end{aligned}$$

Divide both sides by $\bar{F}$ and consider $\bar{\Pi} = 1$, in order to get:

$$\hat{F}_t = (1-\alpha\beta)(-\sigma\hat{C}_t + \hat{Y}_t + \hat{\xi}_t) + \alpha\beta[(\theta-1)E_t\pi_{t+1} + E_t\hat{F}_{t+1}]$$

Use the resource constraint, so that $\hat{C} = \hat{Y}$, to finally obtain:

$$\hat{F}_t = (1-\alpha\beta)[(1-\sigma)\hat{Y}_t + \hat{\xi}_t] + \alpha\beta[(\theta-1)E_t\pi_{t+1} + E_t\hat{F}_{t+1}] \quad (13)$$

Finally, linearize equation (8), by taking into consideration that $\bar{\Pi} = 1$ and that, in the steady state, the equation is: $\frac{\bar{K}}{\bar{F}} = 1$. We obtain:

$$\begin{aligned}
\frac{1}{\bar{F}}dK_t - \frac{\bar{K}}{\bar{F}^2}dF_t &= \frac{1+\theta\phi}{1-\theta}\left(\frac{1-\alpha\bar{\Pi}^{\theta-1}}{1-\alpha}\right)^{\frac{1+\theta\phi}{1-\theta}-1}\left(\frac{-\alpha(\theta-1)\bar{\Pi}^{\theta-1}}{1-\alpha}\right)d\Pi_t \\
\frac{\bar{K}}{\bar{F}}\hat{K}_t - \frac{\bar{K}}{\bar{F}}\hat{F}_t &= \frac{1+\theta\phi}{1-\theta}\left(\frac{1-\alpha\bar{\Pi}^{\theta-1}}{1-\alpha}\right)^{\frac{1+\theta\phi}{1-\theta}-1}\left(\frac{-\alpha(\theta-1)\bar{\Pi}^{\theta}}{1-\alpha}\right)\pi_t \\
\hat{K}_t - \hat{F}_t &= \frac{\alpha}{1-\alpha}(1+\theta\phi)\pi_t \qquad (14)
\end{aligned}$$

To obtain (11), we proceed following two steps. First of all, substitute (12) and (13) into (14):

$$(1-\alpha\beta)[(\sigma+\phi)\hat{Y}_t - (\phi+1)\hat{A}_t] + \alpha\beta[\theta(1+\phi)-(\theta-1)]E_t\pi_{t+1} + \alpha\beta E_t(\hat{K}_{t+1}-\hat{F}_{t+1})$$
$$= \frac{\alpha}{1-\alpha}(1+\theta\phi)\pi_t$$
$$(\phi+\sigma)(1-\alpha\beta)\left(\hat{Y}_t - \frac{1+\phi}{\phi+\sigma}\hat{A}_t\right) + \alpha\beta(1+\theta\phi)E_t\pi_{t+1} + \alpha\beta E_t(\hat{K}_{t+1}-\hat{F}_{t+1})$$
$$= \frac{\alpha}{1-\alpha}(1+\theta\phi)\pi_t$$

Using (**??**):

$$(\phi+\sigma)(1-\alpha\beta)(\hat{Y}_t - \hat{Y}_t^n) + \alpha\beta(1+\theta\phi)E_t\pi_{t+1} + \alpha\beta E_t(\hat{K}_{t+1}-\hat{F}_{t+1}) = \frac{\alpha}{1-\alpha}(1+\theta\phi)\pi_t$$

We can then isolate the term $E_t(\hat{K}_{t+1}-\hat{F}_{t+1})$ to get:

$$E_t(\hat{K}_{t+1}-\hat{F}_{t+1}) = \frac{(1+\theta\phi)}{\beta(1-\alpha)}\pi_t - \frac{(\phi+\sigma)(1-\alpha\beta)}{\alpha\beta})(\hat{Y}_t - \hat{Y}_t^n) - (1+\theta\phi)E_t\pi_{t+1}$$

The second step consists in iterating one period forward equation (14), take the expectation on both sides and substitute the LHS with the equation previously obtained. Formally:

$$\hat{K}_{t+1} - \hat{F}_{t+1} = \frac{\alpha}{1-\alpha}(1+\theta\phi)\pi_{t+1}$$
$$E_t(\hat{K}_{t+1} - \hat{F}_{t+1}) = \frac{\alpha}{1-\alpha}(1+\theta\phi)E_t\pi_{t+1}$$

Substitute the LHS:

$$\frac{(1+\theta\phi)}{\beta(1-\alpha)}\pi_t - \frac{(\phi+\sigma)(1-\alpha\beta)}{\alpha\beta}(\hat{Y}_t - \hat{Y}_t^n) - (1+\theta\phi)E_t\pi_{t+1} = \frac{\alpha}{1-\alpha}(1+\theta\phi)E_t\pi_{t+1}$$

Isolate π_t to get:

$$\pi_t = \frac{(1-\alpha)(1-\alpha\beta)}{\alpha}\frac{\phi+\sigma}{1+\theta\phi}(\hat{Y}_t - \hat{Y}_t^n) + [\beta(1-\alpha)+\alpha\beta]E_t\pi_{t+1}$$

$$\pi_t = \frac{(1-\alpha)(1-\alpha\beta)}{\alpha}\frac{\phi+\sigma}{1+\theta\phi}(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t\pi_{t+1}$$

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t\pi_{t+1}$$